

Drawing extensive form games

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1 Introduction

The goal of this short note is to summarize how to draw extensive form games in `fletcher`. First, we develop a simple drawing style and methodology using basic `fletcher` constructs. After that, we investigate how to add extra elements to extensive form games drawn with `fletcher` in `Typst`.

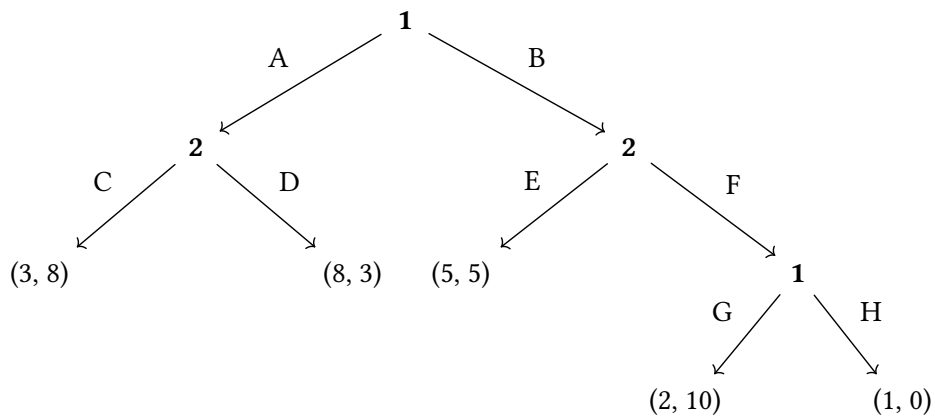
2 Extensive form games with `fletcher`

2.1 Some extra setup

```
1 #let pi-thick-red-arrow = (stroke: 2pt + red,
2   marks: (none, (inherit: ">", size: 1.5)))
3 #let pi-thick-blue-arrow = (stroke: 2pt + blue,
4   marks: (none, (inherit: ">", size: 1.5)))
5 #let pi-label-inside = (label-fill: true, label-anchor: "center")
```

2.2 Examples

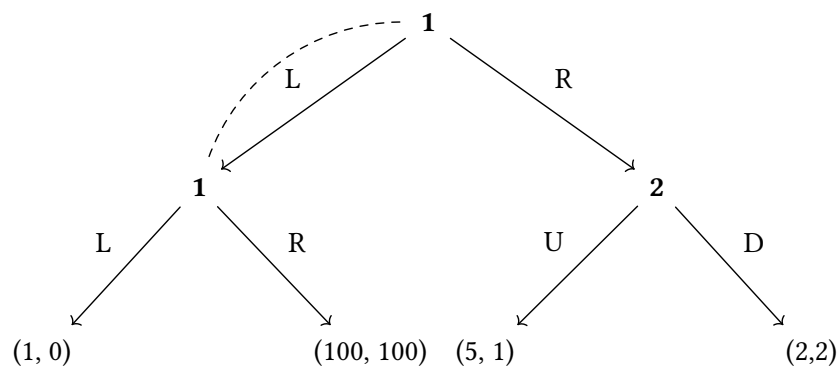
```
1 #diagram(
2   spacing: (20mm, 10mm),
3   node((0,0), [*1*]),
4   edge((-1,1), "->", label: [A]),
5   edge((1,1), "->", label: [B]),
6   node((-1,1), [*2*]),
7   edge((-1.75,2), "->", label: [C]),
8   edge((-0.25,2), "->", label: [D]),
9   node((-1.75,2), [(3, 8)]),
10  node((-0.25,2), [(8, 3)]),
11  node((1,1), [*2*]),
12  edge((0.25, 2), "->", [E]),
13  edge((1.75, 2), "->", [F]),
14  node((0.25,2), [(5, 5)]),
15  node((1.75,2), [*1*]),
16  edge((1.25,3), "->", label: [G]),
17  edge((2.25, 3), "->", label: [H]),
18  node((1.25,3), [(2, 10)]),
19  node((2.25, 3), [(1, 0)])
20 )
```



```

1 #diagram(
2     spacing: (20mm, 15mm),
3     node((0,0), [*1*]),
4     edge((-1,1), "->", label: [L]),
5     edge((1,1), "->", label: [R]),
6     edge((-1,1), "dashed", bend: -35deg),
7     node((-1,1),[*1*]),
8     edge((-1.75,2), "->", label: [L]),
9     edge((-0.25,2), "->", label: [R]),
10    node((-1.75,2), [(1, 0)]),
11    node((-0.25,2), [(100, 100)]),
12    node((1,1), [*2*]),
13    edge((0.25, 2), "->", [U]),
14    edge((1.75, 2), "->", [D]),
15    node((0.25,2), [(5, 1)]),
16    node((1.75,2), [(2,2)]),
17 )

```



```

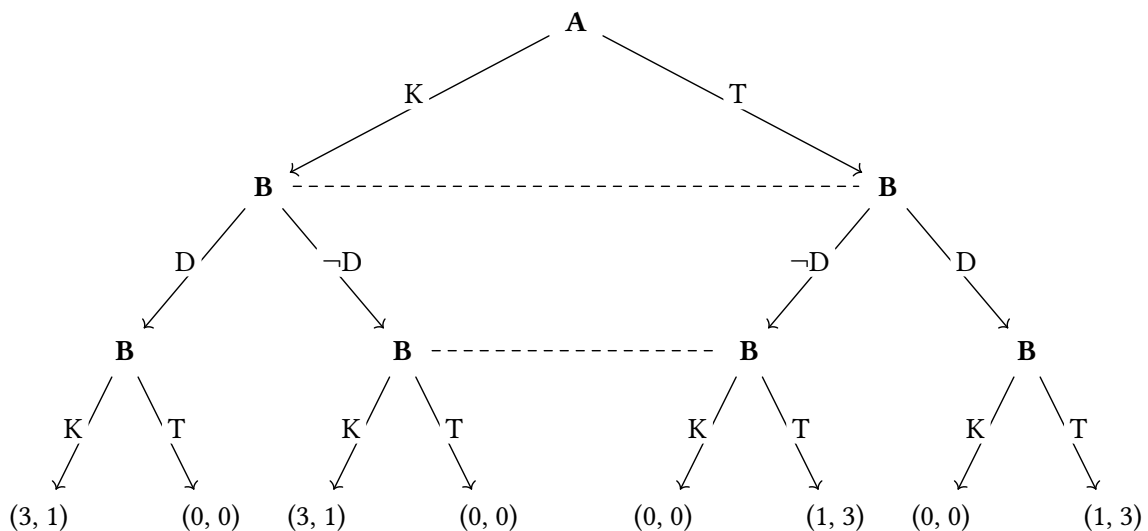
1 #diagram(
2     spacing: (2mm, 15mm),
3     node((0,0), [*A*]),
4     edge((-3.5,1), "->", label: [K], label-fill: true, label-anchor: "center"),
5     edge((3.5,1), "->", label: [T], label-fill: true, label-anchor: "center"),
6     node((-3.5,1),[*B*]),
7     edge((-5,2), "->", label: [D], label-fill: true, label-anchor: "center"),

```

```

8     edge((-2,2), "->", label: [$\text{not}$D], label-fill: true, label-anchor:
      "center"),
9     edge((3.5,1), "dashed"),
10    node((3.5,1), [*B*]),
11    edge((2, 2), "->", [$\text{not}$D], label-fill: true, label-anchor: "center"),
12    edge((5, 2), "->", [D], label-fill: true, label-anchor: "center"),
13    node((-5, 2), [*B*]),
14    edge((-6,3), "->", label: [K], label-fill: true, label-anchor: "center"),
15    edge((-4,3), "->", label: [T], label-fill: true, label-anchor: "center"),
16    node((-2,2), [*B*]),
17    edge((-3,3), "->", label: [K], label-fill: true, label-anchor: "center"),
18    edge((-1,3), "->", label: [T], label-fill: true, label-anchor: "center"),
19    edge((2,2), "dashed"),
20    node((2,2), [*B*]),
21    edge((1, 3), "->", label: [K], label-fill: true, label-anchor: "center"),
22    edge((3, 3), "->", label: [T], label-fill: true, label-anchor: "center"),
23    node((5,2), [*B*]),
24    edge((4, 3), "->", label: [K], label-fill: true, label-anchor: "center"),
25    edge((6, 3), "->", label: [T], label-fill: true, label-anchor: "center"),
26    node((-6,3), [(3, 1)]),
27    node((-4,3), [(0, 0)]),
28    node((-3,3), [(3, 1)]),
29    node((-1,3), [(0, 0)]),
30    node((6,3), [(1, 3)]),
31    node((4,3), [(0, 0)]),
32    node((3,3), [(1, 3)]),
33    node((1,3), [(0, 0)]),
34 )

```



```

1  #diagram(
2      spacing: (8mm, 10mm),
3      node((3.5, 0), [*A*]),
4      edge((0.5, 2), "->", [X], ..pi-label-inside),

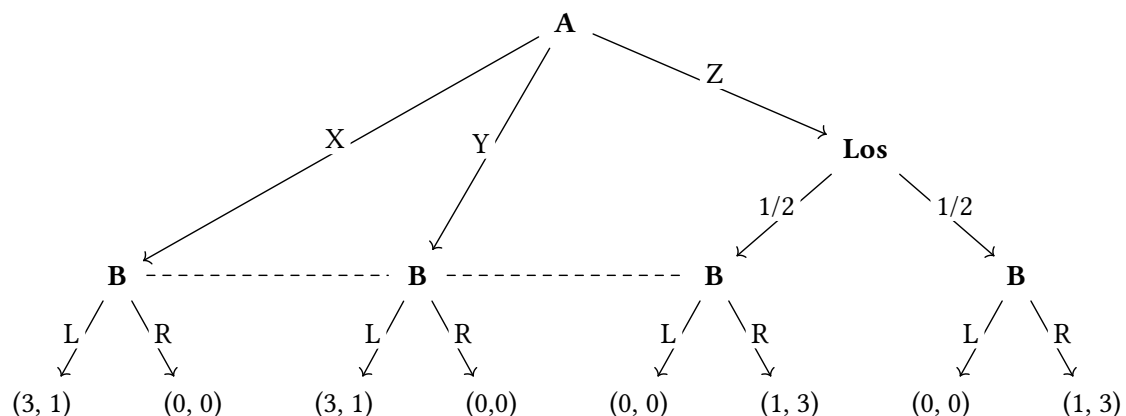
```

t Typst

```

5   edge((2.5, 2), "->", [Y], ..pi-label-inside),
6   edge((5.5, 1), "->", [Z], ..pi-label-inside),
7   node((0.5, 2), [*B*]),
8   edge(((2.5, 2)), "dashed"),
9   edge((0, 3), "->", label: "L", ..pi-label-inside),
10  edge((1, 3), "->", label: "R", ..pi-label-inside),
11  node((0, 3), [(3, 1)]),
12  node((1, 3), [(0, 0)]),
13  node((2.5, 2), [*B*]),
14  edge(((4.5, 2)), "dashed"),
15  edge((2, 3), "->", label: "L", ..pi-label-inside),
16  edge((3, 3), "->", label: "R", ..pi-label-inside),
17  node((2, 3), [(3, 1)]),
18  node((3, 3), [(0, 0)]),
19  node((5.5, 1), [*Los*]),
20  edge((4.5, 2), "->", [1/2], ..pi-label-inside),
21  edge((6.5, 2), "->", [1/2], ..pi-label-inside),
22  node((4.5, 2), [*B*]),
23  edge((4, 3), "->", label: "L", ..pi-label-inside),
24  edge((5, 3), "->", label: "R", ..pi-label-inside),
25  node((4, 3), [(0, 0)]),
26  node((5, 3), [(1, 3)]),
27  node((6.5, 2), [*B*]),
28  edge((6, 3), "->", label: "L", ..pi-label-inside),
29  edge((7, 3), "->", label: "R", ..pi-label-inside),
30  node((6, 3), [(0, 0)]),
31  node((7, 3), [(1, 3)]),
32 )

```



2.2.1 Gra bufetowej (beer-quiche game)

```

1   #diagram(
2       spacing: (10mm, 10mm),
3       node((0, 0), [*Los*]),
4       edge((-2, 0), "->", label: [$p$]),
5       edge((2, 0), "->", label: [$1-p$]),

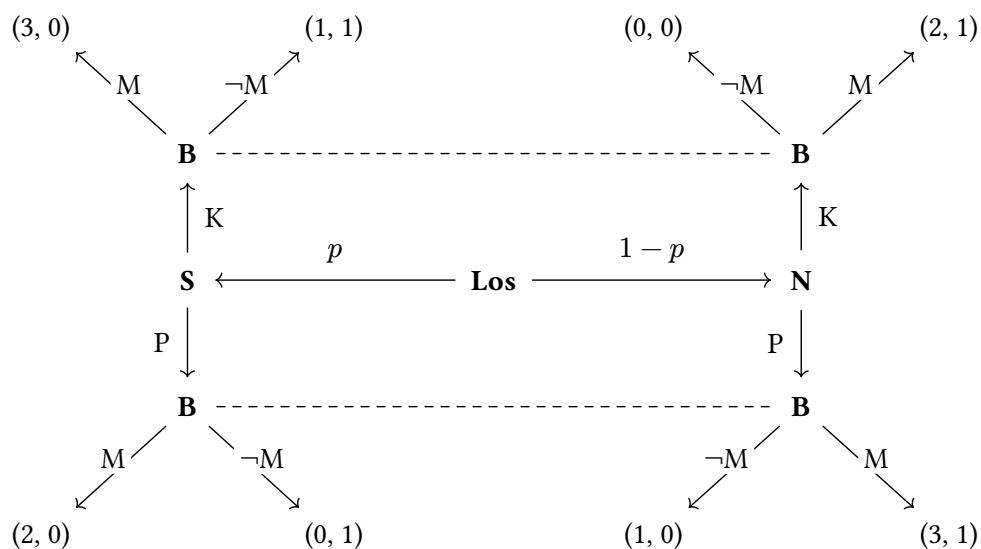
```

tTypst

```

6   node((-2,0), [*S*]),
7   edge((-2,-1), "->", label: [K]),
8   edge((-2,1), "->", label: [P]),
9   node((2,0), [*N*]),
10  edge((2,-1), "->", label: [K]),
11  edge((2,1), "->", label: [P]),
12  node((-2,-1), [*B*]),
13  edge((-3,-2), "->", label: [M], ..pi-label-inside ),
14  edge((-1,-2), "->", label: [$not$M], ..pi-label-inside),
15  edge((2,-1), "dashed"),
16  node((-2,1), [*B*]),
17  edge((-3,2), "->", label: [M], ..pi-label-inside),
18  edge((-1,2), "->", label: [$not$M], ..pi-label-inside),
19  edge((2,1), "dashed"),
20  node((2,-1), [*B*]),
21  edge((3,-2), "->", label: [M], ..pi-label-inside),
22  edge((1,-2), "->", label: [$not$M], ..pi-label-inside),
23  node((2,1), [*B*]),
24  edge((3,2), "->", label: [M], ..pi-label-inside),
25  edge((1,2), "->", label: [$not$M], ..pi-label-inside),
26  node((-3,-2), [(3, 0)]),
27  node((-1,-2), [(1, 1)]),
28  node((1,-2), [(0, 0)]),
29  node((3,-2), [(2, 1)]),
30  node((-3,2), [(2, 0)]),
31  node((-1,2), [(0, 1)]),
32  node((1,2), [(1, 0)]),
33  node((3,2), [(3, 1)]),
34 )

```



2.3 Decorations

```
1 #diagram(
```

tTypst

```

2     spacing: (6mm, 15mm),
3     node((0,0), text(fill: blue)[*A*]),
4     edge((-2,1), ..pi-thick-blue-arrow, label: [K]),
5     edge((2,1), "->", label: [T], ),
6     node((-2,1),text(fill: red)[*B*]),
7     edge((-3,2), ..pi-thick-red-arrow, label: [K]),
8     edge((-1,2), "->", label: [T]),
9     node((2,1), text(fill: red)[*B*]),
10    edge((1,2), "->", label: [K]),
11    edge((3,2), ..pi-thick-red-arrow, label: [T]),
12    node((-3,2), [(3, 1)]),
13    node((-1,2), [(0, 0)]),
14    node((3,2), [(1, 3)]),
15    node((1,2), [(0, 0)]),
16 )

```

